

$$\begin{aligned} & 3 \ln x \Big|_1^2 \\ &= 3(\ln 2 - \ln 1) \\ &= 3 \ln 2 \\ &= \log_e 8 \rightarrow \boxed{B} \end{aligned}$$

$$4^m = 27$$

$$\log_e 4^m = \log_e 3^3$$

$$m = \frac{3 \log_e 3}{2 \log_e 2} \quad \boxed{D}$$

$$\begin{aligned} & 6(e^{3x+2})^5 \times 3 \\ &= 18(e^{3x+2})^5 \quad \boxed{A} \end{aligned}$$

$$y' = \frac{1}{3} x^{-2} \times -1.$$

$$= -\frac{1}{3x^2} \quad \boxed{C}$$

$$y' = 3x^2 - 12x + 12 = 0$$

$$(x-2)^2 = 0$$

$$\Rightarrow x = 2.$$

$$y = 10$$

$$(2, 10) \quad \boxed{C}$$

$$6. \frac{2e^x - xe^x}{(e^x)^2}$$

$$= \frac{2(1-x)}{e^x} \quad \boxed{A}$$

$$7. \frac{1}{2\sqrt{2}} \sec^2(\sqrt{x}-2) \quad \boxed{C}$$

$$8. y' = -\sin x e^{\cos x} \quad \boxed{B}$$

$$9. -\frac{1}{2} \times \frac{(2-x^2)^3}{-3} + C$$

$$\frac{1}{6} (2-x^2)^3 + C \quad \boxed{B}$$

$$10. \quad \boxed{D}$$

Section II

Q11

a) Find the value of $\int_0^1 e^{-\frac{x}{2}} dx$ in exact form

$$\begin{aligned} & \int_0^1 e^{-\frac{x}{2}} dx \\ &= -\frac{1}{\frac{1}{2}} \left[e^{-\frac{x}{2}} \right]_0^1 \\ &= -2 \left[e^{-\frac{1}{2}} - e^0 \right] \quad \text{1 mark for integration} \\ &= -2 \left[e^{-\frac{1}{2}} - 1 \right] \\ &= -2e^{-\frac{1}{2}} + 2 \\ &= 2 - 2(e^{\frac{1}{2}} - 1) \quad \text{1 mark for correct ANS.} \end{aligned}$$

b) $\int_{\frac{\pi}{15}}^{\frac{\pi}{2}} 15 \cos(4x) dx$

$$\begin{aligned} &= \frac{15}{4} \left[\sin 4x \right]_{\frac{\pi}{15}}^{\frac{\pi}{2}} \\ &= \frac{15}{4} \left[\sin 2\pi - \sin \frac{4\pi}{15} \right] \quad \text{1 mark for integration} \\ &= \frac{15}{4} \left[0 - \sin \frac{4\pi}{15} \right] \\ &= -\frac{15}{4} \sin \frac{4\pi}{15} \quad \text{1 mark for correct ANS.} \end{aligned}$$

c) The function $f(x)$ is defined by the rule $f(x) = x^2(6-x)$ in the domain $-1 \leq x \leq 6$.

3

i) Find the stationary points of $f(x)$ and determine their nature.

$$f'(x) = 6x^2 - x^3$$

$$= 12x - 3x^2$$

$$f'(x) = 0 \text{ (stationary point)}$$

$$12x - 3x^2 = 0$$

$$3x(4-x) = 0$$

$$x = 0, x = 4$$

1 mark for diff.

$$\text{at } x=0, y = 0^2(6-0) = 0 \rightarrow (0,0)$$

$$x=4, y = 4^2(6-4) = 32 \rightarrow (4,32)$$

$\therefore (0,0)$ and $(4,32)$ stationary points.

TESTING STATIONARY POINTS

$$f''(x) = 12 - 6x$$

$$\text{at } x=0, f''(0) = 12 > 0$$

$\therefore (0,0)$ minimum.

$$\text{at } x=4, f''(4) = 12 - 24 = -12 < 0$$

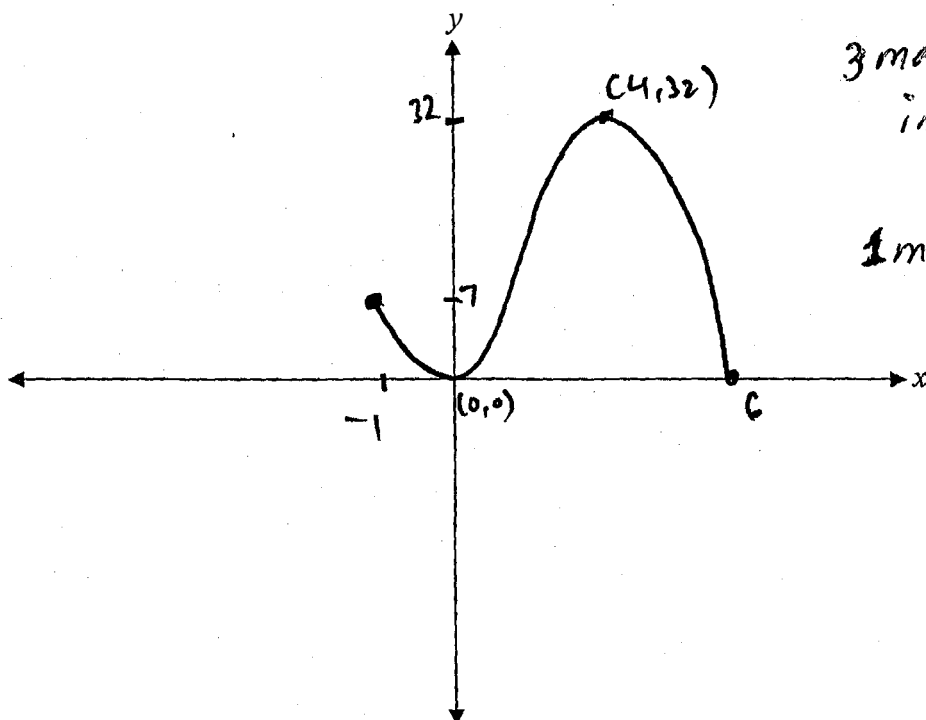
$\therefore (4,32)$ maximum.

1 mark for min, max.

1 mark for stationary points coordinates

3

ii) Sketch the graph of $y = f(x)$. Showing all intercept and stationary points.



3 mark Correct graph include correct labels.

1 mark graphs No label.

iii) State the range of $f(x)$.

at Domain $-1 \leq x \leq 6$

Range $\therefore 0 \leq y \leq 32$.

1 mark Correct Ans

Q12

a) Find $\int \frac{dx}{(3x-1)^2}$

2

$$\int (3x-1)^{-2} dx$$

$$\frac{1}{3} \left[\frac{(3x-1)^{-2+1}}{-2+1} \right] + C$$

2 marks for Correct AN's
Include Simplifying

$$= -\frac{1}{3} (3x-1)^{-1} + C$$

$$= -\frac{1}{3(3x-1)} + C$$

1 mark for just integrator

b) Find $\int \frac{3x}{x^2+1} dx$

2

$$= \frac{3}{2} [\ln(x^2+1)] + C$$

2 marks Correct AN's

1 mark for $\ln(x^2+1)$

c) Find $\int \frac{x^3 - x^2 + 1}{x^2} dx$

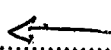
2

$$= \int \frac{x^3 - x^2 + 1}{x^2} dx$$

$$= \int \left(x - 1 + \frac{1}{x^2} \right) dx$$

$$= \int (x - 1 + x^{-2}) dx$$

$$= \frac{x^2}{2} - x + \frac{x^{-2+1}}{-2+1} + C$$



1 mark

$$= \frac{x^2}{2} - x - \frac{1}{x} + C$$

2 marks Correct AN's

d) Find $\int_3^5 \frac{e^x}{e^x - 7} dx$

3

State your answer as an exact value in simple form.

$$= \left[\ln(e^x - 7) \right]_3^5 \quad (1)$$

$$= \ln(e^5 - 7) - \ln(e^3 - 7) = \ln \left[\frac{e^5 - 7}{e^3 - 7} \right] \quad (1)$$

$$= 2.38$$

① identify ln

② integrate correctly

③ substitute & simplify

Q13

a) Differentiate the following $y = \ln \frac{x(x^2 - 1)}{x + 1} = \ln \left[\frac{x(x-1)(x+1)}{(x+1)} \right] = \ln[x(x-1)]$ 2

$$y = \ln[x(x-1)]$$

$$y = \ln x + \ln(x-1)$$

$$\frac{3x^2 - 1}{x^3 - x} = \frac{1}{x+1} + \frac{2x-1}{x(x-1)}$$

$$\frac{dy}{dx} = \frac{1}{x} + \frac{1}{x-1} \quad \left\{ \begin{array}{l} \text{full} \\ \text{marks} \end{array} \right. \quad \leftarrow 1 \text{ mark}$$

$$= \frac{x-1+x}{x(x-1)} = \frac{2x-1}{x(x-1)} = \frac{2x-1}{x^2-x} \quad \leftarrow 2 \text{ marks} \quad \text{Correct ANS}$$

$$\frac{2x^3 + 3x^2 - 1}{x(x+1)^2(x-1)}$$

b) Find the equation of the tangent to the curve $y = x \cos x$ at the point $(\frac{\pi}{3}, \frac{\pi}{6})$ 3

$$\frac{dy}{dx} = x(-\sin x) + \cos x$$

$$\frac{dy}{dx} = -x \sin x + \cos x$$

at $x = \frac{\pi}{3}$, $m_1 = \frac{dy}{dx} = -\frac{\pi}{3} \sin \frac{\pi}{3} + \cos \frac{\pi}{3} = -\frac{\sqrt{3}}{2} \cdot \frac{\pi}{3} + \frac{1}{2}$

$$= -\frac{\sqrt{3}\pi}{6} + \frac{1}{2} = -\frac{\sqrt{3}\pi + 3}{6} \quad \leftarrow 1 \text{ mark for gradient}$$

Equation of tangent

$$y - \frac{\pi}{6} = \left(\frac{-\sqrt{3}\pi + 3}{6} \right) \left(x - \frac{\pi}{3} \right)$$

OR $(\sqrt{3}\pi - 3)x + 6y - \sqrt{3}\pi^2 = 0$ QAO

1 mark for correct equation

$$y = \frac{1}{18} (-3\sqrt{3}\pi x + 9x + \sqrt{3}\pi^2)$$

Q14

a) Use two applications of the Trapezoidal rule to find an approximation to $\int_2^4 \ln x dx$

3

Correct to 3 significant figures. $f(x) = \ln x$

$$h = \frac{4-2}{2} = 1$$

$$f(2) = \ln 2 = 0.693$$

$$f(3) = \ln 3 = 1.097$$

$$f(4) = \ln 4 = 1.386$$

$$\int_2^4 \ln x dx \approx \frac{h}{2} [f(2) + 2f(3) + f(4)]$$

$$= \frac{1}{2} [0.693 + 2(1.097) + 1.386]$$

$$= 2.1365$$

$$= 2.14 \text{ 3sig fig}$$

[3] marks Correct ANS.

[2] marks for Correct (3sig) Integration

[1] mark for correct Applying Trapezoidal Rule

b) Find the $\lim_{x \rightarrow 0} \frac{\sin 4x}{x}$

2

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$4 \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \leftarrow [1] \text{ mark}$$

$$= 4 \times 1 = 4$$

2 mark Correct ANS

Q15 a) Prove that $\tan \theta (\sec \theta - \sin \theta \tan \theta) = \sin \theta$

2

$$\text{LHS} = \frac{\sin \theta}{\cos \theta} \left[\frac{1}{\cos \theta} - \frac{\sin \theta \sin \theta}{\cos \theta} \right]$$

$$= \frac{\sin \theta}{\cos \theta} \left[\frac{1 - \sin^2 \theta}{\cos \theta} \right] \leftarrow [1] \text{ mark}$$

$$= \frac{\sin \theta}{\cos \theta} \left[\frac{\cos^2 \theta}{\cos \theta} \right]$$

$$= \frac{\sin \theta \cos^2 \theta}{\cos \theta}$$

$$= \sin \theta \text{ R.H.S.}$$

[2] mark Correct proof

b) Hence: Solve $2 \tan \theta (\sec \theta - \sin \theta \tan \theta) = 1$ for $0 \leq \theta \leq 2\pi$.

2

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

← [1] mark

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (0 \leq \theta \leq 2\pi) \quad \leftarrow [1] \text{ mark}$$

Q16 Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{2 + \sin x} dx$. State your answer as an exact value in simplest form.

2

$$= \ln [2 + \sin x] \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \ln [2 + \sin \frac{\pi}{2}] - \ln [2 + \sin -\frac{\pi}{2}]$$

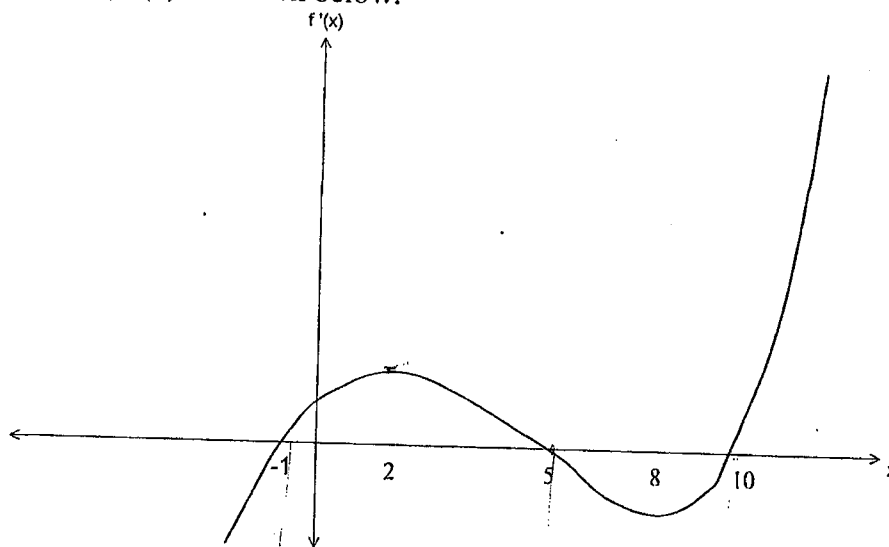
$$= \ln 3 - \ln 1$$

$$= \ln 3$$

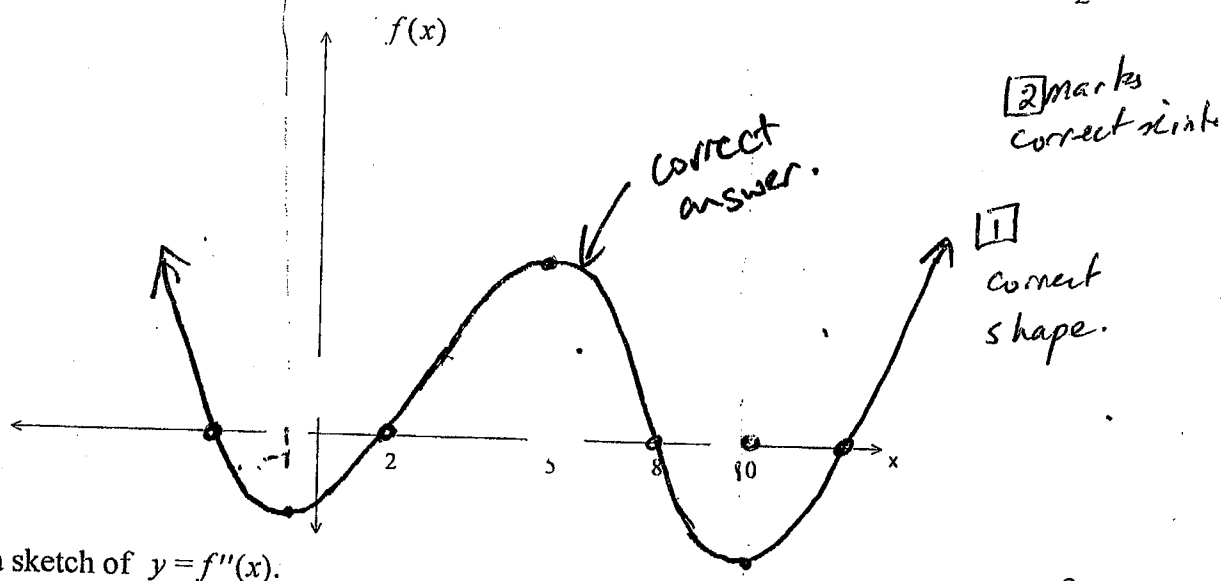
[2] marks Correct ANs

[1] mark for integration

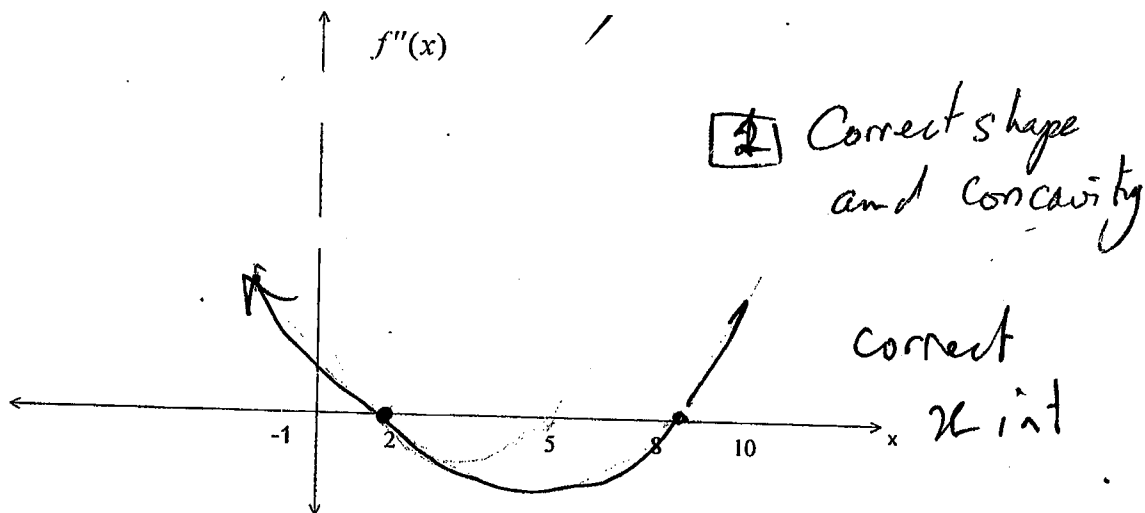
Q1: 7 The graph of $y = f'(x)$ is shown below.



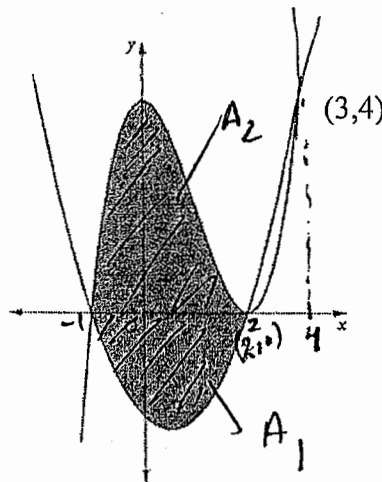
(i) Draw a sketch of $y = f(x)$.



(ii) Draw a sketch of $y = f''(x)$.



Q18 The graphs of $f(x) = x^3 - 3x^2 + 4$ and $g(x) = x^2 - x - 2$ are drawn on the diagram.



a) One of the points of intersection is (3, 4). The other points of intersection lies on the x-axis.

2

Find the coordinates of the other two points of intersection.

$$y = x^3 - 3x^2 + 4 \quad \text{①}$$

$$y = (x^2 - x - 2) = 0$$

$$(x+1)(x-2) = 0$$

← ① mark

$$x = -1, x = 2$$

$$\text{at } x = -1, y = 0 \quad (-1, 0)$$

$$x = 2, y = 0 \quad (2, 0)$$

← ① mark

b) Calculate the shaded area of the region.

5

$$\int_{-1}^2 (x^3 - 3x^2 + 4 - (x^2 - x - 2)) dx$$

$$= \int_{-1}^2 (x^3 - 4x^2 + x + 6) dx$$

$$\left[\frac{x^4}{4} - \frac{4x^3}{3} + \frac{x^2}{2} + 6x \right]_{-1}^2$$

$$\frac{2^4}{4} - \frac{4 \times 2^3}{3} + \frac{2^2}{2} + 6 \times 2 - \left[\frac{(-1)^4}{4} - \frac{4 \times (-1)^3}{3} + \frac{(-1)^2}{2} + 6(-1) \right]$$

$$\frac{16}{4} - \frac{32}{3} + \frac{4}{2} + 12 - \left[\frac{1}{4} + \frac{4}{3} + \frac{1}{2} - 6 \right]$$

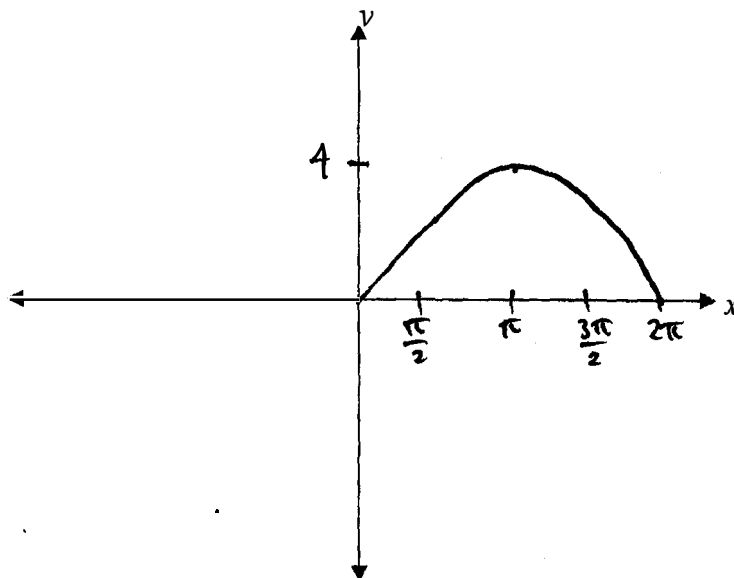
$$4 \times \frac{22}{3} + \frac{47}{12}$$

$$\frac{3+16+16}{12} - \frac{72}{12}$$

$$= \frac{45}{4} u^2. \text{ not } \frac{41}{12}$$

Consider the curve $y = 4 \sin\left(\frac{x}{2}\right)$, $0 \leq x \leq 2\pi$.

a) Sketch the curve.



2

b) Find the area bounded by the curve and the x -axis.

$$A = \int_0^{2\pi} \left(4 \sin \frac{x}{2}\right) dx$$

$$= -4 \times \frac{1}{\frac{1}{2}} \left[\cos\left(\frac{x}{2}\right) \right]_0^{2\pi}$$

$$= -8 [\cos \pi - \cos 0]$$

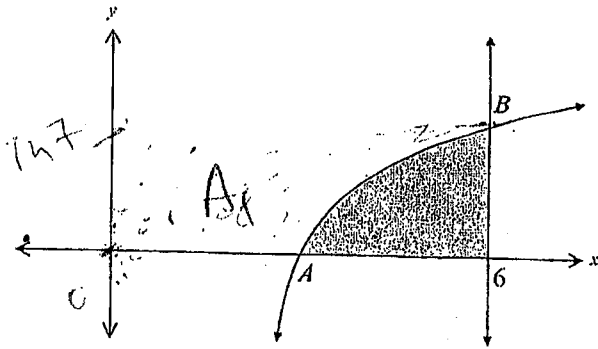
$$= -8 [-1 - 1]$$

$$= -8(-2)$$

$$= 16 \text{ unit}^2$$

Q20 The diagram shows a shaded region which is bounded by the curve $y = \ln(2x - 5)$, the x axis and the line $x = 6$.

The curve $y = \ln(2x - 5)$ intersects the x axis at A and the line $x = 6$ at B .



Req. area =

Ar. of rect.

$\frac{1}{2}$

Area = $\int_0^{\ln 7} \frac{e^y + 5}{2} dy$

a) Show that the coordinates of points A and B are $(3, 0)$ and $(6, \ln 7)$ respectively.

$$y = \ln(2x - 5)$$

$$\text{at } y = 0,$$

$$0 = \ln(2x - 5)$$

$$\ln 1 = \ln(2x - 5)$$

$$1 = 2x - 5$$

$$2x = 6 \rightarrow x = 3 \rightarrow (3, 0)$$

[1 mark for A]

$$\text{at } x = 6$$

$$y = \ln(2x - 5)$$

$$= \ln[2(6) - 5]$$

$$= \ln(12 - 5)$$

$$= \ln 7$$

$$\therefore (6, \ln 7)$$

[1 mark for B]

b) Show that if $y = \ln(2x - 5)$, then $x = \frac{e^y + 5}{2}$.

$$e^y = e^{\ln(2x - 5)}$$

$$e^y = 2x - 5$$

← 1 mark

$$2x = e^y + 5$$

$$x = \frac{e^y + 5}{2}$$

← 1 mark

$$\int_3^6 \frac{1}{x} dx$$

Area of Rect = $6 \ln 7$

$$A_1 = \int_0^{\ln 7} \frac{e^y + 5}{2} dy$$

$$= \left[\frac{e^y + 5y}{2} \right]_0^{\ln 7}$$

$$\frac{1}{2} [e^{\ln 7} + 5 \ln 7 - e^0]$$

$$\frac{1}{2} [6 + 5 \ln 7]$$

$$\therefore \text{Req } A = 6 \ln 7 + 3 - \frac{5}{2} \ln 7$$

$$= \frac{7 \ln 7 + 3}{2}$$

16



3/1/2015
13/12/15

Q21 a) Write the derivative of $\log_e(\tan x)$, $\tan x > 0$.

1

$$= \frac{1}{\tan x} \times \sec^2 x$$

$$= \operatorname{cosec} x \cdot \sec x$$

1 mark

b) Hence find $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec} x \sec x \, dx$

3

Leave your answer as an exact value in simplest form.

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec} x \sec x \, dx$$

$$= \left[\log_e(\tan x) \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

← 1 mark

$$= \log_e\left(\tan \frac{\pi}{3}\right) - \log_e\left(\tan \frac{\pi}{4}\right)$$

$$= \log_e\left(\frac{\sqrt{3}}{1}\right) - \log_e(1)$$

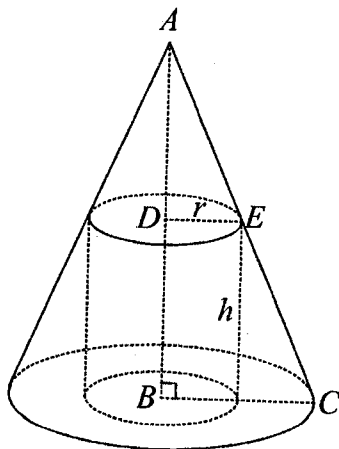
← 1 mark

$$= 0.55$$

← 1 mark

Q22 The radius of the base of a cone is 9 cm and its height is 18 cm.

A cylinder of radius r and height h is inscribed in the cone as shown below.



a) Show that $h = 18 - 2r$.

3

$$\angle ADE = \angle ABC = 90^\circ$$

$\angle AED = \angle ACB$ corresponding angles in parallel lines.

$$\therefore \triangle ADE \sim \triangle ABC$$

$$AB = 18$$

$$AD = 18 - h$$

$$\frac{AD}{AB} = \frac{DE}{BC}$$

$$AD = 18 - h$$

$$\frac{18-h}{18} = \frac{r}{9}$$

$$h = 18 - 2r$$

b) Show that the volumes of the cylinder of maximum volume and the cone,

3

are in the ratio 4:9.

$$V = \pi r^2 h$$

$$= \pi r^2 (18 - 2r)$$

$$= 18\pi r^2 - 2\pi r^3$$

$$\Rightarrow V' = 36\pi r - 6\pi r^2$$

$$= 6\pi r(6 - r)$$

$$V' = 0$$

$$\Rightarrow 6\pi r(6 - r) = 0$$

$$\rightarrow r = 0 \text{ or } 6$$

But r can't be 0.

$$\therefore r = 6$$

1 mark for select $r = 6$ as max.

END OF EXAMINATION



BAULKHAM HILLS HIGH SCHOOL

YEAR 12 TASK 2

2021

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used.
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- A reference sheet is provided

Total marks – 76

This paper consists of TWO sections.

Section I – (10 marks) Pages (2-5)

- Attempt Questions 1-10
- Answer on answer sheet provided
- Allow about 15 minutes for this section

Section II – (66 marks) Pages (7-21)

- Attempt Questions 11-22
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1-10

1. What is the value of $\int_1^2 \frac{3}{x} dx$?

(A) $2 \log_e 3$

(B) $\log_e 8$

(C) $2 \log_e 3^2$

(D) $3 \log_e 3$

2. What is the solution of $4^m = 27$?

(A) $m = \frac{3 \log_e 3}{4}$

(B) $m = \log_e \left(\frac{27}{4} \right)$

(C) $m = \frac{27}{\log_e 4}$

(D) $m = \frac{3 \log_e 3}{2 \log_e 2}$

3. What is the derivative of $(e^{3x+2})^6$?

(A) $18 (e^{3x+2})^6$

(B) $6(e^{3x+2})^5$

(C) $18(e^{3x+2})^7$

(D) $6(e^{3x+2})^6$

4. What is the derivative of $f(x) = \frac{1}{3x}$?

(A) $-\frac{3}{x^2}$

(B) $-\frac{1}{(3x)^2}$

(C) $-\frac{1}{3x^2}$

(D) $\frac{3}{x^2}$

5. What is the stationary point on the curve $y = x^3 - 6x^2 + 12x + 2$?

(A) $(-2, -54)$

(B) $(2, 0)$

(C) $(2, 10)$

(D) $(-2, 0)$

6. What is the derivative of $\frac{2x}{e^x}$?

(A) $\frac{2 - 2x}{e^x}$

(B) $\frac{e^{-x} - xe^{-x}}{x^2}$

(C) $e^{-x}(2 + 2x)$

(D) $\frac{2 + 2x}{e^x}$

7. If $f(x) = \tan(\sqrt{x} - 2)$, what is $f'(x)$?

(A) $2\sec^2(\sqrt{x} - 2)$

(B) $\sec^2(\sqrt{x} - 2)$

(C) $\frac{1}{2\sqrt{x}}\sec^2(\sqrt{x} - 2)$

(D) $\frac{1}{2}\sec^2(\sqrt{x} - 2)$

8. What is the derivative of $f(x) = e^{\cos x}$

(A) $\cos x e^{\cos x}$

(B) $-\sin x e^{\cos x}$

(C) $e^{-\sin x}$

(D) $-\cos x e^{-\sin x}$

9. Which of the following is an expression for $\int \frac{x}{(2 - x^2)^4} dx$?

(A) $\frac{1}{2(2 - x^2)^3} + C$

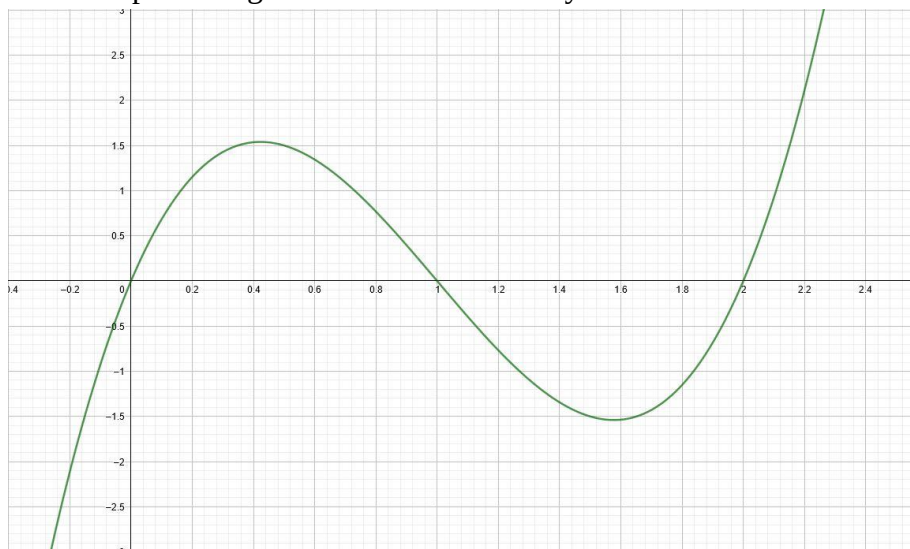
(B) $\frac{1}{6(2 - x^2)^3} + C$

(C) $\frac{1}{4(2 - x^2)^4} + C$

(D) $\frac{-1}{3(2 - x^2)^3} + C$

10. The diagram shows the cubic $y = 4x(x - 1)(x - 2)$.

Which expression gives the area bounded by the cubic and the x -axis?



- (A) $\int_0^2 (4x(x - 1)(x - 2)) dx$
- (B) $\left| \int_2^0 (4x(x - 1)(x - 2)) dx \right|$
- (C) $\left| \int_0^1 (4x(x - 1)(x - 2)) dx \right| + \int_1^2 (4x(x - 1)(x - 2)) dx$
- (D) $\int_0^1 (4x(x - 1)(x - 2)) dx + \left| \int_1^2 (4x(x - 1)(x - 2)) dx \right|$

End of Section I

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BAULKHAM HILLS HIGH SCHOOL

YEAR 12 TASK 2, 2021

MATHEMATICS ADVANCED

**FINAL
MARK**

NESA#: _____

Teacher: _____

Question	Curve-sketching using Calculus	Integration	Exponentials & Logarithmic functions	Trigonometric Functions	Total
MC	4, 5 /2	9, 10 /2	1, 2, 3, 6 /4	7, 8 /2	/10
11	c /7		a /2	b /2	/11
12		a, b, c /6	d /3		/9
13			a /2	b /3	/5
14		a /3		b /2	/5
15				a /4	/4
16				/2	/2
17	/4				/4
18		a, b /5			/5
19				/4	/4
20			a, b, c /7		/7
21				a, b /4	/4
22	a, b /6				/6
Total	/19	/16	/18	/23	/76



BAULKHAM HILLS HIGH SCHOOL

YEAR 12 TASK 2 - 2021

MATHEMATICS ADVANCED

NESA#: _____

Teacher: _____

Mathematics

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$

- (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

- A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

- A ☒ B ☒ C ☐ D ☐

-
- Start Here** → 1. A ☐ B ☐ C ☐ D ☐ 6. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐ 7. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐ 8. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐ 9. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐ 10. A ☐ B ☐ C ☐ D ☐
-

Section II Answer the questions in the spaces provided. Extra writing paper is available if required. Your responses should include relevant mathematical reasoning and/or calculations.

Q11 a) Find the value of $\int_0^1 e^{\frac{-x}{2}} dx$ in exact form.

2

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b) Find the value of $\int_{\frac{\pi}{15}}^{\frac{\pi}{2}} 15 \cos(4x) dx$.

2

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c) The function $f(x)$ is defined by the rule $f(x) = x^2(6 - x)$ in the domain $-1 \leq x \leq 6$.

i) Find the stationary points of $f(x)$ and determine their nature.

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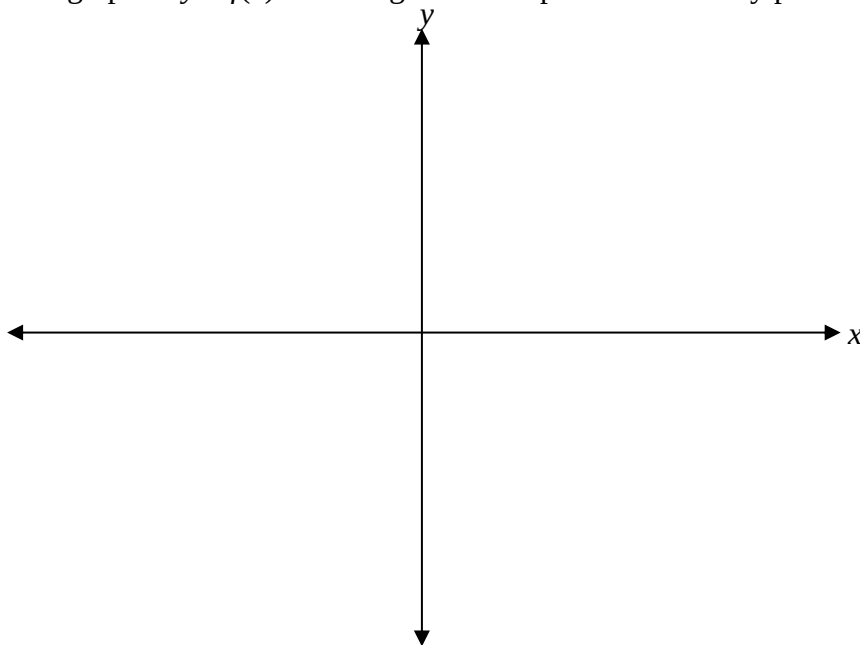
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ii) Sketch the graph of $y = f(x)$. Showing all intercepts and stationary points.

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iii) State the range of $f(x)$.

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Q12 a) Find $\int \frac{dx}{(3x-1)^2}$.

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b) Find $\int \frac{3x}{x^2+1} dx$.

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c) Find $\int \frac{x^3 - x^2 + 1}{x^2} dx$.

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d) Find $\int_3^5 \frac{e^x}{e^x - 7} dx$. State your answer as an exact value in simplest form.

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Q13 a) Differentiate $y = \ln \frac{x(x^2 - 1)}{x + 1}$.

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b) Find the equation of the tangent to the curve $y = x \cos x$ at the point $\left(\frac{\pi}{3}, \frac{\pi}{6}\right)$.

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- Q14 a) Use two applications of the trapezoidal rule to find an approximation to $\int_2^4 \ln x \, dx$. 3
Give your answer correct to 3 significant figures.

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- b) Find $\lim_{x \rightarrow 0} \frac{\sin 4x}{x}$. Show all working. 2

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- Q15 a) Prove that $\tan \theta (\sec \theta - \sin \theta \tan \theta) = \sin \theta$. 2

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b) Hence: Solve $2 \tan \theta (\sec \theta - \sin \theta \tan \theta) = 1$ for $0 \leq \theta \leq 2\pi$.

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Q16 Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{2 + \sin x} dx$. State your answer as an exact value in simplest form.

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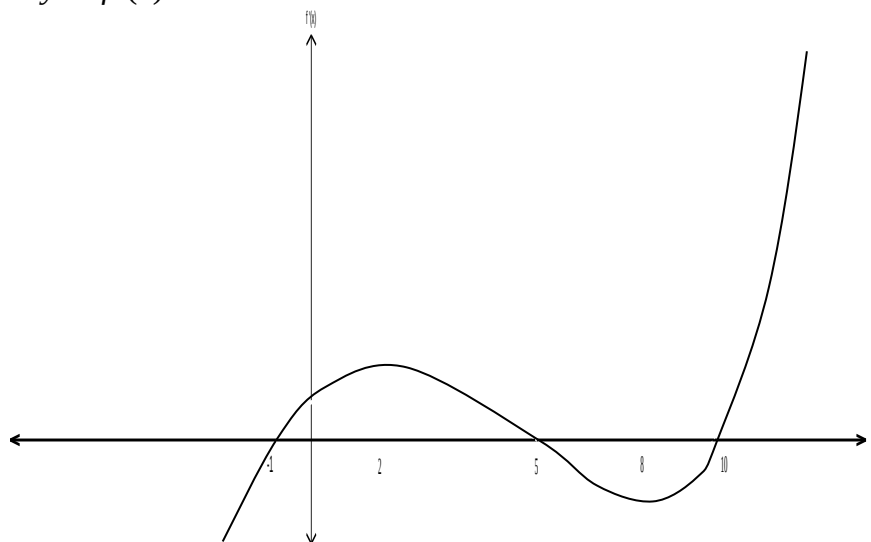
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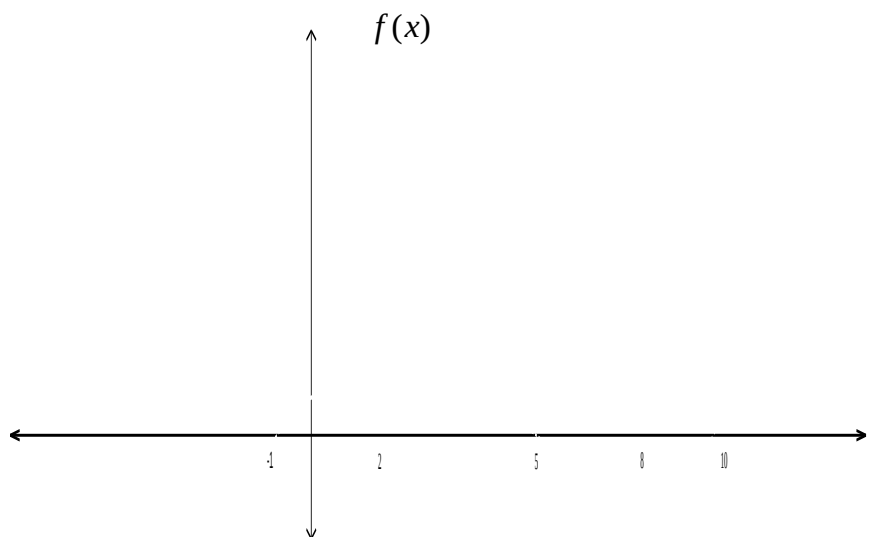
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Q17 The graph of $y = f'(x)$ is shown below.



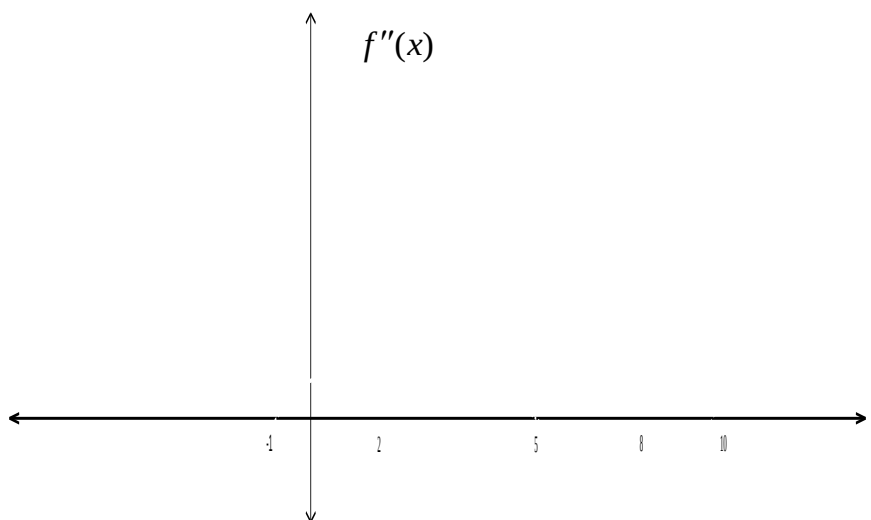
(i) Draw a sketch of $y = f(x)$.

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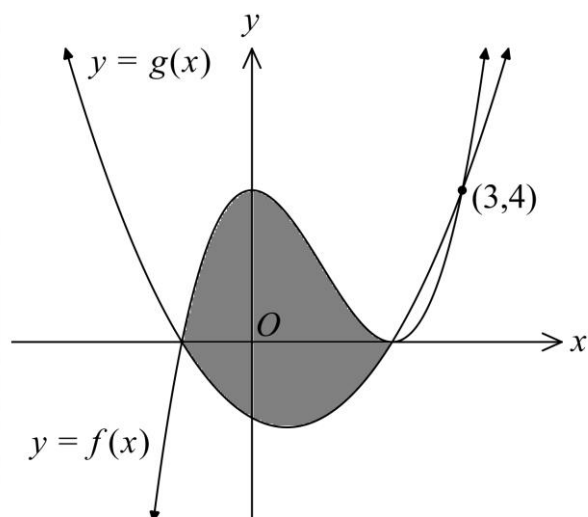


(ii) Draw a sketch of $y = f''(x)$.

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Q18 The graphs of $f(x) = x^3 - 3x^2 + 4$ and $g(x) = x^2 - x - 2$ are drawn on the diagram.



- a) One of the points of intersection is $(3, 4)$. The other points of intersection are on the x -axis. Find the coordinates of the other two points of intersection.

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- b) Calculate the shaded area of the region.

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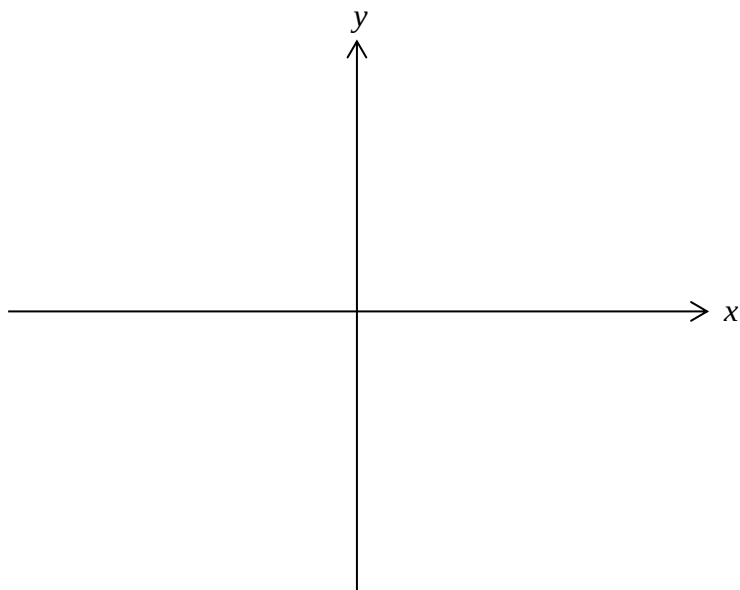
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Q19 Consider the curve $y = 4\sin\left(\frac{x}{2}\right), 0 \leq x \leq 2\pi$.

a) Sketch the curve.

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b) Find the area bounded by the curve and the x -axis.

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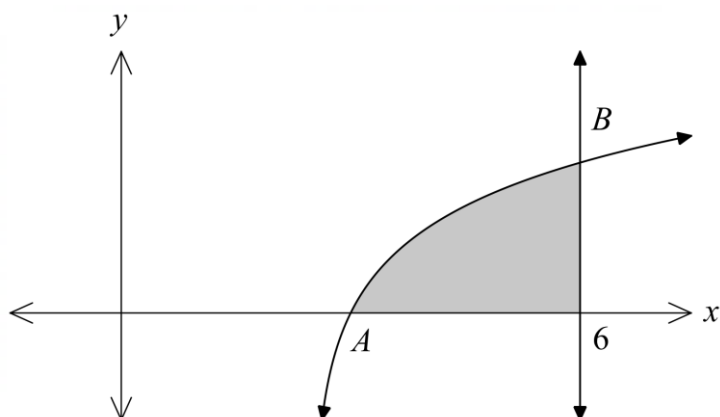
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Q20 The diagram shows a shaded region which is bounded by the curve $y = \ln(2x - 5)$, the x -axis and the line $x = 6$.

The curve $y = \ln(2x - 5)$ intersects the x axis at A and the line $x = 6$ at B .



a) Show that the coordinates of points A and B are $(3, 0)$ and $(6, \ln 7)$ respectively.

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b) Show that if $y = \ln(2x-5)$, then $x = \frac{e^y + 5}{2}$.

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c) Hence find the exact area of the shaded region.

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Q21 a) Write the derivative of $\log_e(\tan x)$, where $\tan x > 0$.

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b) Hence find $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \operatorname{cosec} x \sec x \, dx$.

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Leave your answer as an exact value in simplest form.

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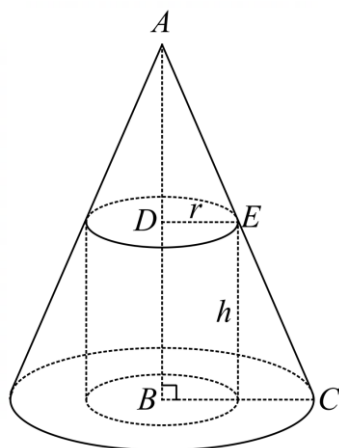
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Q22 The radius of the base of a cone is 9 cm and its height is 18 cm.

A cylinder of radius r and height h is inscribed in the cone as shown below.



a) Show that $h = 18 - 2r$.

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b) Show that the volumes of the cylinder of maximum volume and the cone,
are in the ratio 4:9.

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END OF EXAMINATION